

```

1 1WA24CS226
2 #include <stdio.h>
3 #include <math.h>
4
5 int gcd(int a, int b) {
6     return b ? gcd(b, a % b) : a;
7 }
8
9 int lcm(int a, int b) {
10     return (a * b) / gcd(a, b);
11 }
12
13 int main() {
14     int n, i, hp;
15     float util = 0;
16
17     printf("Enter the number of processes: ");
18     scanf("%d", &n);
19
20     int C[n], P[n];
21
22     printf("Enter CPU burst times:\n");
23     for (i = 0; i < n; i++)
24         scanf("%d", &C[i]);
25
26     printf("Enter the time periods:\n");
27     for (i = 0; i < n; i++)
28         scanf("%d", &P[i]);
29
30     hp = P[0];
31     for (i = 1; i < n; i++)
32         hp = lcm(hp, P[i]);
33
34     printf("LCM=%d\n\n", hp);
35
36     printf("Rate Monotone Scheduling:\n");
37     printf("PID\tBurst\tPeriod\n");
38
39     for (i = 0; i < n; i++) {
40         printf("%d\t%d\t%d\n", i + 1, C[i], P[i]);
41         util += (float)C[i] / P[i];
42     }

```

```
44 float bound = n * (pow(2, 1.0/n) - 1);
45 printf("\n%f <= %f -> %s\n", util, bound,
46       (util <= bound) ? "true" : "false");
47
48 printf("Scheduling occurs for %d ms\n\n", hp);
49
50 int time = 0;
51 while (time < hp) {
52     for (i = 0; i < n; i++) {
53         if (time % P[i] == 0) {
54             printf("%dms onwards: Process %d running\n", time, i + 1);
55             time += C[i];
56             break;
57         }
58     }
59     time++;
60 }
61
62 return 0;
63 }
64
```

Enter the number of processes: 2

Enter CPU burst times:

20 35

Enter the time periods:

50 100

LCM=100

Rate Monotone Scheduling:

PID	Burst	Period
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1	20	50
---	----	----

2	35	100
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$0.750000 \leq 0.828427 \rightarrow \text{true}$

Scheduling occurs for 100 ms

0ms onwards: Process 1 running

50ms onwards: Process 1 running

Process returned 0 (0x0) execution time : 25.895 s

Press any key to continue.